NEW LOWER BOUNDS FOR ERDŐS–SZEKERES NUMBERS IN THREE AND FOUR DIMENSIONS

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ABSTRACT. *We construct eleven points in general position in four-dimensional space with no eight of them in convex position, and fifteen points in general position in three-dimensional space with no eight of them in convex position. The first set raises the lower bound for the least number of points in four-space that forces eight in convex position from eleven to twelve, so that this number is now known to be twelve or thirteen. The second raises the corresponding lower bound in three-space from fourteen to sixteen. For nine points we construct thirteen points in four-space and eighteen points in three-space with no nine in convex position, which raise the lower bounds from thirteen to fourteen and from fifteen to nineteen. All four sets have small integer coordinates, and each proof is a finite list of identities that place one point of the set strictly inside a simplex spanned by others.*

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1. INTRODUCTION

A finite set of points in $\mathbb{R}^d$ is in *general position* if no $d+1$ of its points lie on a common hyperplane, and in *convex position* if none of its points lies in the convex hull of the others. A set of k points in convex position is a *k-gon*; a k-gon of a set X is a *k-hole* of X if its convex hull contains no other point of X.

In 1935 Erdős and Szekeres [1] proved that for every k there is a least integer $g^{(2)}(k)$ such that every set of $g^{(2)}(k)$ points in general position in the plane contains a k-gon.[1] The value $g^{(2)}(5) = 9$ goes back to Makai [2]. Szekeres and Peters proved $g^{(2)}(6) = 17$ with a computer search [3]; Marić gave a formally verified proof [4], and Scheucher a proof by satisfiability solving [7]. In 2017 Suk proved $g^{(2)}(k) \le 2^{k+o(k)}$ [5], so the conjecture holds up

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[1]The question came from Esther Klein, who had observed that five points in general position in the plane always contain the vertices of a convex quadrilateral; she married Szekeres soon after, and the result became known as the happy ending problem. The paper [1] contains two proofs, one through Ramsey's theorem and one through cups and caps, and the second gives $g^{(2)}(k) \le \binom{2k-4}{k-2} + 1$. Erdős and Szekeres conjectured $g^{(2)}(k) = 2^{k-2} + 1$ and later constructed 2^{k-2} points without a k-gon, which shows that the conjectured value cannot be lowered. The survey of Morris and Soltan [2] tells this history and collects the improvements of the upper bound up to 2000.

TABLE 1. Values and bounds of $g^{(d)}(k)$ for $3 \le d \le 5$ and $5 \le k \le 9$. Entries in bold are proved in this paper; the earlier lower bounds were 14 and 15 for $d = 3$, and 11 and 13 for $d = 4$. The upper bounds 13 are Scheucher's [7].

	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$
$d=3$	6	9	13	$\ge$ **16**	$\ge$ **19**
$d=4$	5	7	9	**12**–13	$\ge$ **14**
$d=5$	5	6	8	10	12–13

to the error term in the exponent. For holes the answer changes: there are arbitrarily large sets in general position in the plane without 7-holes [2], while Heule and Scheucher proved that every 30 points in general position contain a 6-hole and that 29 points do not suffice [6].

The theorem holds in every dimension. Let $g^{(d)}(k)$ be the least integer such that every set of $g^{(d)}(k)$ points in general position in $\mathbb{R}^d$ contains a k-gon, and let $h^{(d)}(k)$ be the corresponding number for k-holes, possibly infinite. Projection onto a generic plane shows $g^{(d)}(k) \le g^{(2)}(k)$ [2], and Károlyi's dimension reduction [8] gives the sharper estimate $g^{(d)}(k) \le g^{(d-1)}(k-1)+1$. Károlyi and Valtr [9] constructed sets showing $g^{(d)}(k) \ge c_d^{k^{1/(d-1)}}$ for a constant $c_d > 1$, and Pohoata and Zakharov [10] proved $g^{(d)}(k) = 2^{o(k)}$ for every $d \ge 3$, so in three and more dimensions the growth is not even known up to the exponent. Valtr [11] showed that sets without large holes exist in every dimension.

The small values form a short list (Table 1). Trivially $g^{(d)}(k) = k$ for $k \le d+1$, and $g^{(d)}(d+2) = d+3$ by a theorem of Motzkin recorded by Danzer, Grünbaum and Klee [12]; see also [17]. For $d+2 \le k \le \lfloor 3d/2 \rfloor + 1$,

$$g^{(d)}(k) = 2k - d - 1, \tag{1}$$

where the lower bound is a construction of Bisztriczky and Harborth [13], the upper bound is due to Bisztriczky and Soltan [14], and the two are combined in [2, Theorem 4.1]. Beyond this range Bisztriczky and Soltan proved $g^{(3)}(6) = 9$ [14]. Morris and Soltan observed that all values known in 2000 with $k > \lfloor (3d+1)/2 \rfloor$ satisfy $g^{(d)}(k) = 4g^{(d)}(k-d) - 3$ and conjectured this for all such k [2, Conjecture 4.2]. In 2023 Scheucher [7] encoded the problem as a satisfiability question for acyclic chirotopes, the combinatorial abstraction of point sets, and proved

$$g^{(3)}(7) = 13, \qquad g^{(4)}(8) \le 13, \qquad g^{(5)}(9) \le 13, \tag{2}$$

together with the same upper bounds for holes. His twelve points in $\mathbb{R}^3$ without a 7-gon are integer points. His twelve-element witnesses in rank five and six are chirotopes that he proved to be non-realizable by points, and he conjectured that realizable witnesses exist. So the lower bounds $g^{(4)}(8) \ge 11$ and $g^{(5)}(9) \ge 12$ given by (1) remained in place, and in $\mathbb{R}^3$ the only lower bound for $g^{(3)}(8)$ was $g^{(3)}(7)+1 = 14$.[2]

We prove the following three theorems.

Theorem A. *There is a set of eleven points in general position in $\mathbb{R}^4$ with no eight points in convex position. Consequently $g^{(4)}(8) \ge 12$ and $h^{(4)}(8) \ge 12$, and by (2) both numbers lie in $\{12, 13\}$.*

Theorem B. *There is a set of fifteen points in general position in $\mathbb{R}^3$ with no eight points in convex position. Consequently $g^{(3)}(8) \ge 16$ and $h^{(3)}(8) \ge 16$.*

Theorem C. *There is a set of thirteen points in general position in $\mathbb{R}^4$ with no nine points in convex position, and a set of eighteen points in general position in $\mathbb{R}^3$ with no nine points in convex position. Consequently $g^{(4)}(9) \ge 14$, $h^{(4)}(9) \ge 14$, $g^{(3)}(9) \ge 19$, $h^{(3)}(9) \ge 19$ and $g^{(3)}(10) \ge 20$.*

Theorem A is the first lower bound beyond (1) in four dimensions, and it leaves two candidates for $g^{(4)}(8)$; the conjecture of Morris and Soltan and Scheucher's conjecture both predict 13. Theorem B raises the lower bound for $g^{(3)}(8)$ by two; the conjecture of Morris and Soltan predicts $g^{(3)}(8) = 4g^{(3)}(5) - 3 = 21$, and the best upper bound we know comes from dimension reduction and the planar estimate of Tóth and Valtr, as in [7]. Theorem C

[2] If X has no k-gon and y is any point with $X \cup \{y\}$ in general position, then $X \cup \{y\}$ has no $(k+1)$-gon, because removing y from a $(k+1)$-gon leaves a k-gon of X. Hence $g^{(d)}(k+1) \ge g^{(d)}(k)+1$. The same argument applies to holes, since the convex hull of a subset of a hole contains no point of the set outside the hole.

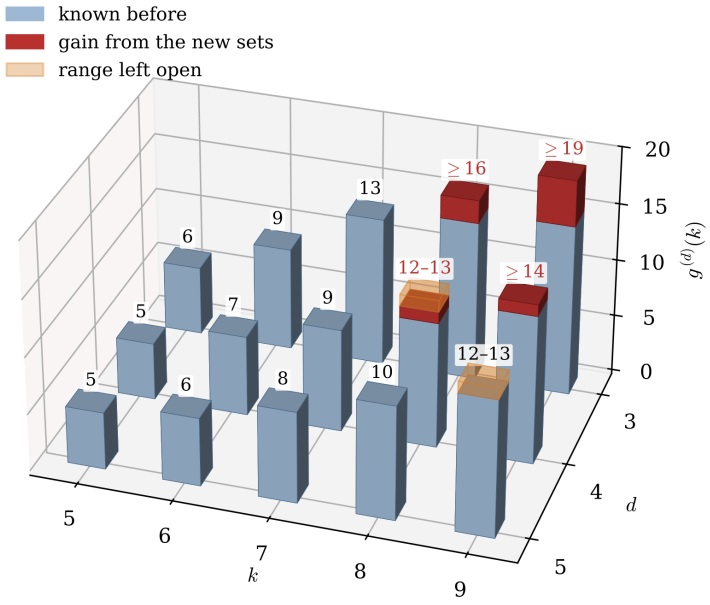


FIGURE 1. The values of $g^{(d)}(k)$ for $d = 3,4,5$ and $k = 5,\dots,9$. Blue bars are the values or lower bounds known before; red bars show the gain from Theorems A, B and C; the light orange blocks mark the ranges left open below Scheucher's upper bounds. Where no upper bound is drawn, only the lower bound is shown.

raises the lower bound for $g^{(4)}(9)$ from the value $2k-d-1=13$ of the construction of Bisztriczky and Harborth to 14, and the lower bound for $g^{(3)}(9)$ from $g^{(3)}(7)+2=15$ to 19; the conjecture of Morris and Soltan predicts $g^{(4)}(9)=4g^{(4)}(5)-3=17$ and $g^{(3)}(9)=4g^{(3)}(6)-3=33$.[3]

All the proofs are certificates. If $v_0,\dots,v_d$ are affinely independent points of $\mathbb{R}^d$ and

$$\sum_{j=0}^{d} m_j v_j = Mp, \qquad m_j \in \{1,2,3,\dots\}, \quad M=\sum_{j=0}^{d} m_j, \tag{3}$$

then p lies in the interior of the simplex spanned by the v_j, and no set containing p and the v_j is in convex position. We call (3) a *relation*. A set has no k-gon as soon as every k of its points contain the $d+2$ points of some relation (Proposition 2.5). Twenty-eight relations prove Theorem A, 161 prove Theorem B, and 46 and 282 prove the two parts of Theorem C; they are listed in Appendix B. Each relation is an identity between integer vectors that can be checked by hand. That the relations meet every 8-subset is a finite combinatorial condition: for Theorem A it is a statement about 165 triples, which Table 6 lets the reader check directly, and for Theorem B it concerns 6435 subsets and is checked by the programs of the archive. For the thirteen points of Theorem C it concerns the 715 four-element complements of 9-subsets, again listed through Table 9, and for the eighteen points the 48620 subsets of size nine. Since relations survive small perturbations (Lemma 2.4), general position costs nothing; the integer sets we give are in general position as they stand.

The two sets have a layered structure (Figures 3 and 4). The eleven points in $\mathbb{R}^4$ consist of the six vertices of a bipyramid over a tetrahedron and five points inside it, and every relation places one of the five inner points in a simplex. The fifteen points in $\mathbb{R}^3$ form three convex layers, a triangular bipyramid, an octahedron and a tetrahedron, and no relation places a vertex of the outer bipyramid inside a simplex. Scheucher's twelve points in $\mathbb{R}^3$ without a 7-gon have convex layers of sizes 5, 6 and 1, again a triangular bipyramid and an octahedron. The sets of Theorem C are built the same way (Figures 5 and 6). The thirteen points in $\mathbb{R}^4$ are the seven vertices of a 4-simplex with two caps and six points inside it forming a cyclic polytope. The eighteen points in $\mathbb{R}^3$ form a capped octahedron, a pentagonal bipyramid and a tetrahedron, and again only points of the two inner layers lie

[3]For $g^{(3)}(10)$ the construction of Bisztriczky and Harborth gives 16, and Theorem C with the monotonicity of the previous footnote gives 20. In $\mathbb{R}^4$ the bound $g^{(4)}(10)\geq g^{(4)}(9)+1=15$ coincides with $2k-d-1=15$, so Theorem C gives nothing new there.

inside simplices. For other extremal problems on explicit configurations in dimensions four and seven see [18, 19], and for reflexive polytopes in dimension four see [20].

Section 2 proves the certificate criterion, Sections 3, 4 and 5 prove Theorems A, B and C, Section 6 explains how the sets were found, and Section 7 discusses what remains open. Appendix A contains the detailed proofs of the lemmas of Section 2, and Appendix B the four certificates.

2. Relations and certificates

Throughout, X is a finite set in $\mathbb{R}^d$ and conv, int denote the convex hull and the interior. For points $v_0,\dots,v_d$ write $\det(v_0,\dots,v_d)$ for the determinant of the $(d+1)\times(d+1)$ matrix whose rows are $(1,v_j)$; it vanishes exactly when the points lie on a common hyperplane, and its absolute value is $d!$ times the volume of the simplex they span.

Lemma 2.1. *Let X be in general position and $Y\subseteq X$. Then Y is in convex position if and only if no point of Y lies in $\operatorname{int}\operatorname{conv}\sigma$ for a set σ of $d+1$ other points of Y.*

Proof. If $p\in\operatorname{int}\operatorname{conv}\sigma$ with $\sigma\subseteq Y\setminus\{p\}$, then $p\in\operatorname{conv}(Y\setminus\{p\})$ and Y is not in convex position. Conversely, suppose $p\in\operatorname{conv}(Y\setminus\{p\})$. By Carathéodory's theorem [17] p is a convex combination of affinely independent points $v_0,\dots,v_r$ of $Y\setminus\{p\}$, and we choose r minimal, so all coefficients are positive. If $r<d$, then p and $v_0,\dots,v_r$ are $r+2\le d+1$ points on an r-dimensional affine subspace, hence on a hyperplane, contrary to general position. So $r=d$, the simplex $\operatorname{conv}\{v_0,\dots,v_d\}$ is full-dimensional, and a point with positive barycentric coordinates lies in its interior. □

Definition 2.2. A *relation* of X is a triple (p,σ,m) where $p\in X$, $\sigma=\{v_0,\dots,v_d\}\subseteq X\setminus\{p\}$ satisfies $\det(v_0,\dots,v_d)\neq 0$, and $m=(m_0,\dots,m_d)$ consists of positive integers with $\sum_j m_j v_j = Mp$, $M=\sum_j m_j$. Its *support* is $\{p\}\cup\sigma$, a set of $d+2$ points.

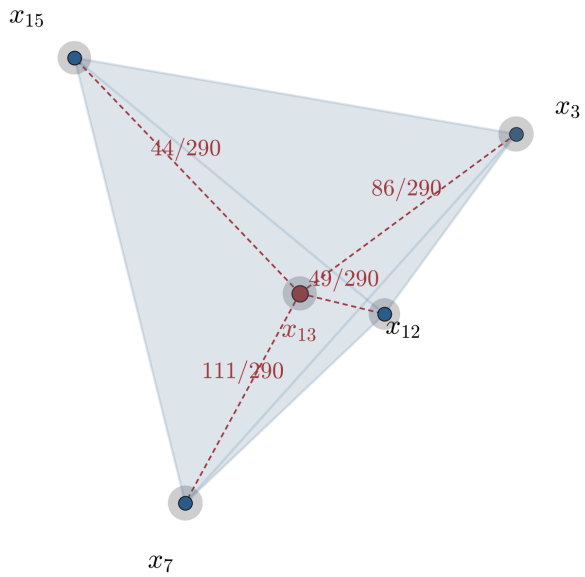


Figure 2. One relation of the set of Theorem B, drawn after an affine map that turns the simplex into a regular tetrahedron and keeps barycentric coordinates. The point p is the weighted mean of the four vertices with the positive integer weights shown. The small balls indicate that every point may move a little without p leaving the simplex (Lemma 2.4).

Lemma 2.3. *If (p,σ,m) is a relation, then $p \in \operatorname{int}\operatorname{conv}\sigma$, the barycentric coordinates of p with respect to σ are m_j/M, and if the points have integer coordinates and $\gcd(m_0,\dots,m_d)=1$, then M divides $\det(v_0,\dots,v_d)$.*

Proof. The v_j are affinely independent, so every point has unique barycentric coordinates with respect to them, and (3) shows that those of p are $m_j/M>0$. Write $D=\det(v_0,\dots,v_d)$ and let N_j be the same determinant with v_j replaced by p. By Cramer's rule the jth barycentric coordinate of p is N_j/D (Appendix A), so $m_jD=MN_j$ for every j. The right-hand sides are integer multiples of M, and an integer combination of the m_j equals 1, hence M divides $\det(v_0,\dots,v_d)$. □

In the tables of the appendices we record, for each relation, the integer $g=|\det(v_0,\dots,v_d)|/M$. A reader who checks one determinant per relation thereby also checks $\det\neq 0$; Lemma 2.3 explains why g is always an integer.

Lemma 2.4. *Let (p,σ,m) be a relation of X. There is $\varepsilon>0$ such that for every map $x\mapsto x'$ with $|x'-x|<\varepsilon$ on the support, p' lies in the interior of the simplex spanned by the images of σ.*

Proof. The barycentric coordinates of p with respect to σ are rational functions of the coordinates of the $d+2$ points whose common denominator is $\det(v_0,\dots,v_d)$. They are therefore continuous where this determinant does not vanish, and at the given points they are positive. Continuity gives an ε that keeps the determinant nonzero and the coordinates positive. Appendix A gives an explicit ε. □

Proposition 2.5. *Let X be a finite set in $\mathbb{R}^d$ and $\mathscr{R}$ a set of relations of X such that every k-subset of X contains the support of some relation in $\mathscr{R}$. Then X contains no k points in convex position. Moreover, there is a set X' in general position, obtained from X by moving each point by an arbitrarily small amount, with the same property.*

Proof. Let $Y\subseteq X$ have k points and let (p,σ,m) be a relation with support in Y. By Lemma 2.3, $p\in\operatorname{conv}\sigma\subseteq\operatorname{conv}(Y\setminus\{p\})$, so Y is not in convex position. For the second statement let ε be the least of the constants of Lemma 2.4 over the finitely many relations of $\mathscr{R}$. The configurations of $|X|$ points in $\mathbb{R}^d$ that are not in general position form the zero set of the product of the polynomials $\det(x_{i_0},\dots,x_{i_d})$, none of which vanishes identically, so they form a closed set with empty interior in $\mathbb{R}^{d|X|}$. Hence there are points x' with $|x'-x|<\varepsilon$ in general position. Every relation of $\mathscr{R}$ still places p' in the interior of the image simplex, so the first argument applies to X'. □

The condition of Proposition 2.5 can be stated through complements. A k-subset Y contains the support S of a relation if and only if $X\setminus Y\subseteq X\setminus S$. So if $|X|=n$, the sets $X\setminus S$ of size $n-d-2$ must cover every $(n-k)$-subset of X. This is a covering design, and Schönheim's bound [15] gives a lower bound for the number of relations any such certificate needs.

Remark 2.6. A certificate in the sense of Proposition 2.5 for eleven points in $\mathbb{R}^4$ and $k=8$ covers the 3-subsets of an 11-set by 5-sets, and needs at least $\lceil\frac{11}{5}\lceil\frac{10}{4}\lceil\frac{9}{3}\rceil\rceil\rceil=18$ relations. For fifteen points in $\mathbb{R}^3$ and $k=8$ it covers the 7-subsets of a 15-set by 10-sets, and needs at least 74 relations. For $k=9$ the thirteen points in $\mathbb{R}^4$ need at least 26 relations and the eighteen points in $\mathbb{R}^3$ at least 85. Our certificates use 28, 161, 46 and 282 relations.

By Lemma 2.1 the method is complete: if a set in general position has no k-gon, then every k-subset contains a point in the interior of a simplex of other points of the subset, and when the coordinates are rational the barycentric coordinates are rational, so the relation can be written with integer weights. A certificate exists exactly when the set has no k-gon.

3. Eleven points in four-space

Let $X_4=\{x_1,\dots,x_{11}\}\subset\mathbb{R}^4$ be the points of Table 2. All coordinates lie between 0 and 40.

Proposition 3.1. *The convex hull of X_4 is a bipyramid over a tetrahedron: its vertices are $x_1,\dots,x_6$, and its eight facets are the tetrahedra spanned by x_1 or x_6 together with three of x_2,x_3,x_4,x_5. The points $x_7,\dots,x_{11}$ lie in the interior.*

Proof. For each of the eight listed quadruples τ the determinants $\det(\tau,x)$ have one sign for all seven remaining points x of X_4, which is a sign check of 56 determinants. So the hyperplane through τ supports $\operatorname{conv}X_4$, and since τ consists of four affinely independent points it spans a facet. Each triangle contained in one of these eight tetrahedra lies in exactly two of them. In a convex polytope every ridge lies in exactly two facets and the facets are connected through ridges [16], so these eight tetrahedra are all the facets of $\operatorname{conv}X_4$, and they are the facets of the bipyramid

TABLE 2. The eleven points of Theorem A in $\mathbb{R}^4$.

point	x	y	z	w
x_1	0	19	34	30
x_2	2	36	25	0
x_3	18	0	19	1
x_4	19	18	40	32
x_5	20	32	0	23
x_6	23	31	15	1
x_7	12	16	24	14
x_8	13	27	21	13
x_9	16	24	14	17
x_{10}	16	25	19	25
x_{11}	18	24	20	14

conv$\{x_1,\dots,x_6\}$. The points $x_7,\dots,x_{11}$ lie strictly on the inner side of all eight facet hyperplanes, by the same sign check, so they are interior. □

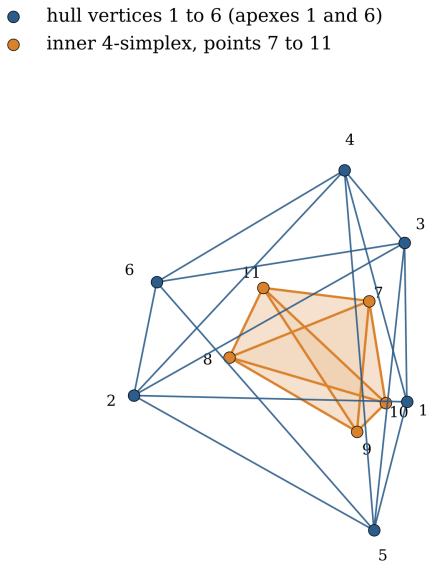


FIGURE 3. A schematic view of the set of Theorem A, projected to three dimensions after an affine map that makes the hull a regular bipyramid. The tetrahedron $x_2x_3x_4x_5$ and the apexes x_1, x_6 span the hull; the five inner points (red) form a 4-simplex, drawn enlarged. Every one of the 28 relations of Table 6 places a red point inside a simplex of other points.

Proof of Theorem A. Each row of Table 6 is a relation of X_4: the identity $\sum_j m_j x_{s_j} = Mx_p$ holds in each of the four coordinates, the weights are positive, and the determinant of the simplex is $\pm gM \neq 0$. The last column of the table lists, for each relation, the five points outside its support. Every one of the $\binom{11}{3} = 165$ triples of indices lies in one of these 28 five-element sets. So the complement of every 8-subset of X_4 lies outside the support of some relation, that is, every 8-subset contains a support, and Proposition 2.5 shows that X_4 has no 8 points in convex position. The set X_4 is itself in general position: the $\binom{11}{5} = 462$ determinants of five of its points are all nonzero, the smallest in absolute value being 11. Independently of this check, Proposition 2.5 provides a perturbation of X_4

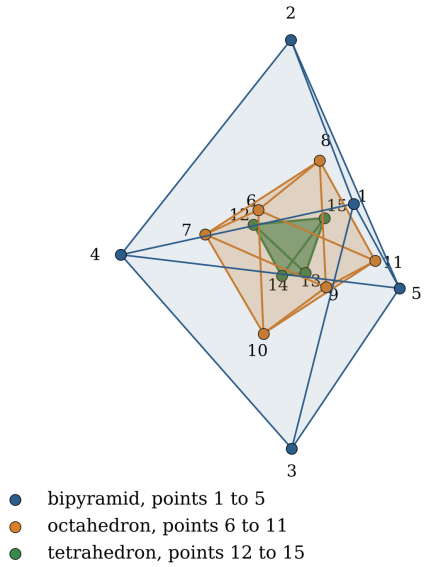


FIGURE 4. The three convex layers of the set of Theorem B: a triangular bipyramid (blue), an octahedron (green) and a tetrahedron (red), with the labels of Table 3. The drawing is schematic: it shows regular solids with the true combinatorics and labels, since in the actual coordinates the inner layers are small and flat.

in general position with the same property. Hence $g^{(4)}(8) \geq 12$. An 8-hole is an 8-gon, so $h^{(4)}(8) \geq g^{(4)}(8) \geq 12$, and Scheucher's bound $h^{(4)}(8) \leq 13$ [7] together with $g^{(4)}(8) \leq h^{(4)}(8)$ gives the last statement. □

Every relation places one of the inner points $x_7, \ldots, x_{11}$ in a simplex; the outer points are vertices of the hull and lie inside nothing. Any eight points of X_4 include at least two inner points, and the certificate shows that one of the inner points among them always lies inside a simplex spanned by four or five of the others.

4. FIFTEEN POINTS IN THREE-SPACE

Let $X_3 = \{x_1, \ldots, x_{15}\} \subset \mathbb{R}^3$ be the points of Table 3. All coordinates lie between 0 and 180.

Proposition 4.1. *The set X_3 has three convex layers. The vertices of* $\operatorname{conv} X_3$ *are $x_1, \ldots, x_5$, forming a triangular bipyramid with apexes x_2 and x_3 over the triangle $x_1x_4x_5$. The vertices of the convex hull of the remaining points are $x_6, \ldots, x_{11}$, forming an octahedron whose opposite vertices are x_6, x_9, x_7, x_{11} and x_8, x_{10}. The points $x_{12}, \ldots, x_{15}$ span a tetrahedron inside it.*

Proof. As for Proposition 3.1. The six triangles of the bipyramid have constant sign of $\det(\tau, x)$ over the other twelve points of X_3, so they are facets of $\operatorname{conv} X_3$; every edge among them lies in exactly two, so they are all the facets, and the remaining ten points lie strictly inside. The eight triangles of the octahedron have constant sign over the other seven points of $\{x_6, \ldots, x_{15}\}$ and close up in the same way, so they are the facets of the second layer, with $x_{12}, \ldots, x_{15}$ strictly inside. Four points in general position span a tetrahedron. □

Proof of Theorem B. Each row of Tables 7 and 8 is a relation of X_3, by the same three checks as before. Every one of the $\binom{15}{8} = 6435$ eight-point subsets of X_3 contains the support of one of these 161 relations; equivalently, every 7-subset of the indices lies in the complement of a support. This finite statement is verified in the archive [22] by independent programs in four languages and by the Lean 4 kernel; three of the programs also test every 8-subset directly for convex position in exact integer arithmetic, without using the certificate. Proposition 2.5 then shows that X_3 has no 8 points in convex position. The set X_3 is in general position, the smallest absolute value of its $\binom{15}{4} = 1365$ determinants being 6, and again Proposition 2.5 would supply a perturbation if this check were omitted. So $g^{(3)}(8) \geq 16$ and $h^{(3)}(8) \geq 16$, and the monotonicity $g^{(3)}(9) \geq g^{(3)}(8) + 1$ gives $g^{(3)}(9) \geq 17$. □

TABLE 3. The fifteen points of Theorem B in $\mathbb{R}^3$.

point	x	y	z
x_1	0	128	0
x_2	53	0	130
x_3	92	166	46
x_4	110	180	108
x_5	171	139	79
x_6	73	88	95
x_7	75	161	71
x_8	79	143	55
x_9	98	169	67
x_{10}	110	176	104
x_{11}	123	128	82
x_{12}	81	109	93
x_{13}	85	154	67
x_{14}	86	154	66
x_{15}	101	163	69

The points p of the relations run through all ten points of the two inner layers, and no relation places a vertex of the outer bipyramid inside a simplex. In 90 of the 161 relations the point p is a vertex of the inner tetrahedron. The others are needed because an 8-subset may avoid the inner tetrahedron altogether: it then consists of eight of the eleven points of the two outer layers, and one of its octahedron vertices must lie inside a simplex of the others.

5. NINE POINTS

Let $Y_4 = \{x_1,\dots,x_{13}\} \subset \mathbb{R}^4$ be the points of Table 4 and $Y_3 = \{x_1,\dots,x_{18}\} \subset \mathbb{R}^3$ those of Table 5; within this section the labels x_i refer to these two tables. All coordinates lie between 0 and 50 in $\mathbb{R}^4$ and between 0 and 150 in $\mathbb{R}^3$.

TABLE 4. The thirteen points of Theorem C in $\mathbb{R}^4$.

point	x	y	z	w
x_1	0	2	46	23
x_2	2	19	7	50
x_3	10	0	0	0
x_4	27	20	42	23
x_5	36	1	26	30
x_6	39	21	23	15
x_7	47	11	21	26
x_8	12	9	8	23
x_9	26	12	19	29
x_{10}	27	10	26	22
x_{11}	30	9	25	25
x_{12}	30	11	20	18
x_{13}	30	16	26	21

Proposition 5.1. *The vertices of* $\operatorname{conv} Y_4$ *are* $x_1,\dots,x_7$*, and its thirteen facets are* $x_2x_3x_4x_6$*, the six tetrahedra* $x_1\tau$ *with* τ *one of the triangles* $x_2x_3x_4$*,* $x_2x_3x_5$*,* $x_2x_4x_5$*,* $x_3x_4x_6$*,* $x_3x_5x_6$*,* $x_4x_5x_6$*, and the six tetrahedra* $x_7\tau$ *with* τ *one of* $x_2x_3x_5$*,* $x_2x_3x_6$*,* $x_2x_4x_5$*,* $x_2x_4x_6$*,* $x_3x_5x_6$*,* $x_4x_5x_6$*. So* $\operatorname{conv} Y_4$ *arises from the* 4*-simplex* $\operatorname{conv}\{x_2,\dots,x_6\}$ *by placing* x_1 *beyond its facets* $x_2x_3x_4x_5$ *and* $x_3x_4x_5x_6$ *and then* x_7 *beyond its facets* $x_2x_3x_5x_6$ *and* $x_2x_4x_5x_6$*, and every two*

TABLE 5. The eighteen points of Theorem C in $\mathbb{R}^3$.

point	x	y	z
x_1	0	150	13
x_2	15	90	8
x_3	17	88	133
x_4	60	0	67
x_5	93	39	2
x_6	96	50	80
x_7	107	143	0
x_8	45	82	28
x_9	50	92	25
x_{10}	59	47	60
x_{11}	60	36	56
x_{12}	69	59	18
x_{13}	77	53	17
x_{14}	79	59	56
x_{15}	59	67	33
x_{16}	62	64	32
x_{17}	70	62	26
x_{18}	73	64	50

vertices except x_1, x_7 span an edge. The points $x_8, \dots, x_{13}$ lie in the interior and are the vertices of a polytope combinatorially equivalent to the cyclic polytope $C_4(6)$.

Proof. The sign check of Proposition 3.1 applies to the thirteen listed tetrahedra: for each of them the determinants $\det(\tau, x)$ have one sign over the other nine points of Y_4, and the remaining six points lie strictly on the inner side of all thirteen hyperplanes. Every triangle of one of the thirteen tetrahedra lies in exactly two of them, so they are all the facets. Placing a point beyond a set of facets of a polytope, and beneath the others, replaces the visible facets by the cones from the point over the boundary of their union [17, Theorem 5.2.1]; the two visible facets of $\operatorname{conv}\{x_2, \dots, x_6\}$ for x_1 share the triangle $x_3x_4x_5$, the two for x_7 share $x_2x_5x_6$, and the facets listed are exactly those this rule produces. The convex hull of $x_8, \dots, x_{13}$ has nine facets,

$$x_8x_9x_{10}x_{11},\ x_8x_9x_{10}x_{13},\ x_8x_9x_{11}x_{12},\ x_8x_9x_{12}x_{13},\ x_8x_{10}x_{11}x_{12},$$
$$x_8x_{10}x_{12}x_{13},\ x_9x_{10}x_{11}x_{13},\ x_9x_{11}x_{12}x_{13},\ x_{10}x_{11}x_{12}x_{13},$$

by the same check, and every two of its six vertices span an edge. A neighbourly simplicial 4-polytope with six vertices is combinatorially the cyclic polytope [17]. □

Proposition 5.2. *The set Y_3 has three convex layers. The first, with vertices $x_1, \dots, x_7$, is the octahedron with vertices $x_2, \dots, x_7$ and opposite pairs $\{x_2, x_6\}$, $\{x_3, x_5\}$, $\{x_4, x_7\}$, capped by x_1 over its face $x_2x_3x_7$. The second, with vertices $x_8, \dots, x_{14}$, is a pentagonal bipyramid with apexes x_9 and x_{11} over the pentagon $x_8x_{10}x_{14}x_{13}x_{12}$. The points $x_{15}, \dots, x_{18}$ span a tetrahedron inside it.*

Proof. The ten triangles $x_1x_2x_3$, $x_1x_2x_7$, $x_1x_3x_7$, $x_2x_3x_4$, $x_2x_4x_5$, $x_2x_5x_7$, $x_3x_4x_6$, $x_3x_6x_7$, $x_4x_5x_6$, $x_5x_6x_7$ have constant sign of $\det(\tau, x)$ over the other fifteen points, every edge among them lies in exactly two, and the eleven points $x_8, \dots, x_{18}$ lie strictly inside; removing x_1 leaves the eight faces of the octahedron, including $x_2x_3x_7$. The ten triangles $x_9x_ix_j$ and $x_{11}x_ix_j$ with x_ix_j an edge of the pentagon pass the same test over the other points of $\{x_8, \dots, x_{18}\}$, with $x_{15}, \dots, x_{18}$ strictly inside. □

Proof of Theorem C. Each row of Table 9 is a relation of Y_4, by the three checks of the proof of Theorem A. Every one of the $\binom{13}{4} = 715$ four-element subsets of the indices lies in one of the 46 seven-element sets of the last column, so every 9-subset of Y_4 contains the support of a relation, and Y_4 has no 9 points in convex position by Proposition 2.5. Its $\binom{13}{5} = 1287$ determinants are nonzero, the smallest in absolute value being 21. Each row of

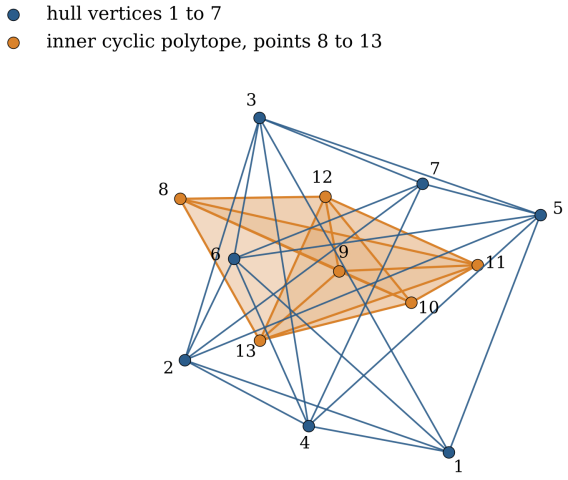


FIGURE 5. The thirteen points of Theorem C in $\mathbb{R}^4$, after an affine map that makes the hull round and an orthogonal projection to $\mathbb{R}^3$ chosen to keep the points apart; the six inner points are drawn enlarged about their centre. Blue lines are the edges of the hull, which contain every pair of vertices except x_1, x_7; the orange polytope is the inner cyclic polytope $C_4(6)$.

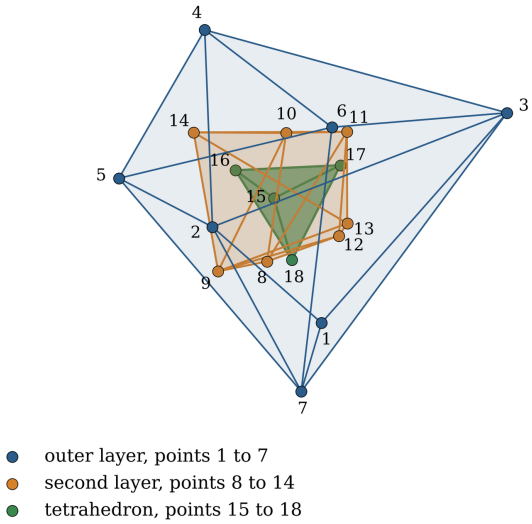


FIGURE 6. The three convex layers of the eighteen points of Theorem C in $\mathbb{R}^3$: a capped octahedron (blue), a pentagonal bipyramid (orange) and a tetrahedron (green). Each layer is drawn after its own affine map and the layers are nested about a common centre, so the picture shows the true combinatorics and labels but not the true sizes.

Tables 10 to 12 is a relation of Y_3, and every one of the $\binom{18}{9} = 48620$ subsets of size nine contains the support of one of these 282 relations; this is verified as for Theorem B, together with a direct test of every 9-subset for convex position. The $\binom{18}{4} = 3060$ determinants of Y_3 are nonzero, the smallest being 18. Hence $g^{(4)}(9) \geq 14$ and $g^{(3)}(9) \geq 19$; holes are gons, so the same bounds hold for $h^{(4)}(9)$ and $h^{(3)}(9)$, and monotonicity gives $g^{(3)}(10) \geq 20$. □

As before, the outer layer takes no part as the point p: every relation of Y_4 places one of $x_8, \ldots, x_{13}$ inside a simplex, and every relation of Y_3 one of $x_8, \ldots, x_{18}$, a vertex of the tetrahedron in 149 of the 282 relations. A 9-subset of Y_4 contains at least two of the six points of the cyclic polytope, and a 9-subset of Y_3 at least two points of the two inner layers.

6. How the sets were found

The sets were found by local search over coordinates. For a candidate set we counted its k-subsets in convex position, but a bare count gives the search nothing to follow, since most moves change it by zero. We therefore scored each k-subset Y by how far it is from failing: for every point p of Y and every simplex σ of other points of Y, the least barycentric coordinate of p with respect to σ measures how far p is from being inside, and the score of Y is f of the smallest such violation over p and σ, with $f(t) = t/(1+t)$ for violations and 0 for subsets that already fail. The total score is zero exactly for sets without k-gons. Simulated annealing on this score, with occasional *repair moves* that take a point of a surviving k-gon and place it just inside the simplex of the subset nearest to it, found the eleven points in $\mathbb{R}^4$ and the fifteen points in $\mathbb{R}^3$, and for $k = 9$ the thirteen points in $\mathbb{R}^4$ and the eighteen points in $\mathbb{R}^3$.[4]

The sets so found have real coordinates and some relations with very small barycentric coordinates. We then *polished* each set by hill climbing on the least barycentric coordinate over a fixed certificate, which doubled it for the set in $\mathbb{R}^3$, and rounded to the coarsest integer grid on which a certificate survives; this gave coordinates at most 40 and 50 for the two sets in $\mathbb{R}^4$, and at most 180 and 150 for the two sets in $\mathbb{R}^3$. Finally we chose a certificate with few relations and small weights by a randomised greedy covering followed by removal of redundant relations. None of these steps is part of the proof.

7. What remains open

The value of $g^{(4)}(8)$ is now 12 or 13. The same search, run on twelve points in $\mathbb{R}^4$, ended with sets that still contain four 8-gons, and twelve points in $\mathbb{R}^5$ without a 9-gon ended at three 9-gons; for fifteen points in $\mathbb{R}^3$ the search reached zero, while sixteen points ended at four 8-gons. Scheucher's twelve-element chirotopes without 8-gons and 9-gons in rank five and six are not realizable either. A twelve-point set in $\mathbb{R}^4$ without an 8-gon would prove $g^{(4)}(8) = 13$ and confirm both conjectures quoted in the introduction; a proof that every twelve points contain an 8-gon would refute them.

In three dimensions the gap is wider: $16 \leq g^{(3)}(8)$, while the conjectured value is 21. The layered structure of X_3 suggests adding a fourth layer inside the tetrahedron or replacing the tetrahedron by a larger polytope. The value of $h^{(3)}(8)$ is not even known to be finite [7].

For nine points the bounds are $14 \leq g^{(4)}(9)$ and $19 \leq g^{(3)}(9)$, against the conjectured 17 and 33. The search, run on fourteen points in $\mathbb{R}^4$ and on nineteen points in $\mathbb{R}^3$, ended with sets that still contain 4 and 12 subsets of nine points in convex position, and adding one point to the sets of Theorem C at any of the 123450 and 51591 sampled positions left at least nine and eight such subsets.

Finally, the certificates are longer than the bounds of Remark 2.6: 28 relations against 18 and 46 against 26 in $\mathbb{R}^4$, and 161 against 74 and 282 against 85 in $\mathbb{R}^3$. In both sets the points p of the relations are confined to the inner layers, so the supports are not spread evenly over the set, as a covering design meeting Schönheim's bound would require.

[4]Before these constructions we checked the status of the problem in the literature cited here and with the BLLM engine Anaxia [21]. The engine keeps indexes of DeepMind's formal-conjectures collection and AlphaEvolve problem repository, of the OpenRefute archive and of OpenAI's collection of mathematical manuscripts, and answers a query with the matching entries and their recorded status. For the Erdős–Szekeres problem these indexes hold only the planar question (Erdős problem 107 in the formal-conjectures collection and problem 58 of the AlphaEvolve repository) and no entry on $g^{(d)}(k)$ for $d \geq 3$, which agreed with our reading of [7] and of the papers citing it. The point sets of this paper were found and verified by the search and certificate programs of the archive [22], not by the engine. The methods that found them, a graded count of violated subsets, repair moves that push a point just inside its nearest simplex, and the polishing of the least barycentric coordinate described below, were afterwards added to the engine as construction rules.

DATA AND CODE AVAILABILITY

The coordinates, the certificates, the search, polishing and rounding programs, the independent checks in Python, C, C++ and Julia, the Lean 4 file that checks all four certificates in the kernel, the figure programs and the LaTeX source are in the repository `github.com/creelie/Erdos-s1` and its Zenodo archive, doi:10.5281/zenodo.23483598 [22]. The status check of the problem was made with the BLLM engine Anaxia, release v3.0.5 [21]. Deep Bhattacharjee carried out the computations and the Python, C, C++, Julia, Lean and shell verifications with AI assistance (Claude, Anthropic).

ETHICAL DECLARATIONS

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AUTHOR CONTRIBUTIONS

Deep Bhattacharjee proposed the problem, carried out the searches and computations and wrote the paper. Ushashi Bhattacharya checked the relations of Tables 6 and 9 and the proofs of Appendix A. Both authors approved the final version.

APPENDIX A. PROOFS OF THE LEMMAS

Barycentric coordinates. Let $v_0,\dots,v_d \in \mathbb{R}^d$ with $D = \det(v_0,\dots,v_d) \neq 0$. For a point p the system $\sum_j \lambda_j = 1$, $\sum_j \lambda_j v_j = p$ has the matrix whose columns are $(1, v_j)$, the transpose of the matrix defining D, so it has the unique solution

$$\lambda_j = \frac{\det(v_0,\dots,v_{j-1},p,v_{j+1},\dots,v_d)}{D} \tag{4}$$

by Cramer's rule. The point p lies in the interior of $\operatorname{conv}\{v_0,\dots,v_d\}$ if and only if all $\lambda_j > 0$: the simplex is the set where all $\lambda_j \geq 0$, and its boundary is where some $\lambda_j = 0$. Summing (4) gives $\sum_j \det(v_0,\dots,p,\dots,v_d) = D$, an identity used to check tables: the numerators of a relation are $m_j D/M$ and add up to D.

Proof of the divisibility in Lemma 2.3. From $\lambda_j = m_j/M$ and (4), $m_j D = M N_j$ with N_j the integer numerator. Choose integers a_j with $\sum_j a_j m_j = 1$. Then $D = \sum_j a_j m_j D = M \sum_j a_j N_j$.

An explicit ε in Lemma 2.4. Let all $d+2$ points lie in a ball of radius $R \geq 1$ around the origin and let $\delta = \min_j m_j/M$. The determinant $\det(v_0,\dots,v_d)$ is a multilinear function of the rows $(1, v_j)$. Moving each point by at most $\varepsilon \leq R$ changes each row by at most ε and keeps every row of norm at most $\sqrt{1+4R^2} \leq 3R$. Replacing the rows one at a time and applying Hadamard's inequality to each difference, the determinant changes by at most $(d+1)\,\varepsilon\,(3R)^d$, and the same holds for each numerator in (4). If $\varepsilon \leq |D|\,\delta/\big(4(d+1)(3R)^d\big)$, then the new determinant has the sign of D and modulus at least $\frac{3}{4}|D|$, each new numerator differs from the old one $\lambda_j D$ by at most $\frac{1}{4}\delta|D|$, and so every new barycentric coordinate is at least $\frac{3}{4}\delta/\frac{5}{4} = \frac{3}{5}\delta > 0$.

The general position claim of Proposition 2.5. Each $\det(x_{i_0},\dots,x_{i_d})$ is a polynomial in the $d|X|$ coordinates that does not vanish identically, since it is nonzero on any set of $d+1$ affinely independent points. A finite union of zero sets of nonzero polynomials has empty interior, so every neighbourhood of a configuration contains configurations in general position.

Schönheim's bound. A family of s-subsets of an n-set covering every t-subset has at least

$$\Big\lceil \frac{n}{s} \Big\lceil \frac{n-1}{s-1} \cdots \Big\lceil \frac{n-t+1}{s-t+1} \Big\rceil \cdots \Big\rceil \Big\rceil$$

members [15]: the members containing a fixed point, with that point removed, cover all $(t-1)$-subsets of the remaining $n-1$ points, and counting incidences gives the recursion. For $(n,s,t) = (11,5,3)$ the bound is $\lceil \frac{11}{5} \lceil \frac{10}{4} \cdot 3 \rceil \rceil = \lceil \frac{11\cdot 8}{5} \rceil = 18$; for $(15,10,7)$ the nested values are $3,6,11,19,31,49,74$. For $k = 9$ the thirteen points in $\mathbb{R}^4$ give $(n,s,t) = (13,7,4)$, with nested values $3,7,14,26$, and the eighteen points in $\mathbb{R}^3$ give $(18,13,9)$, with nested values $2,4,7,12,19,29,43,61,85$.

TABLE 6. The twenty-eight relations behind Theorem A. In each row the point x_p satisfies $\sum_j m_j x_{s_j} = M x_p$ with $\sigma = \{s_1, \ldots, s_5\}$ listed in increasing order, the weights m_j are positive integers with sum M, and $|\det \sigma| = gM$ for the 5×5 determinant of the rows $(1, x_s)$, $s \in \sigma$. The last column lists the five points not used by the relation; every triple of $\{1, \ldots, 11\}$ lies in one of these sets.

p	σ	m_1	m_2	m_3	m_4	m_5	M	g	unused points
7	1,2,3,4,10	7673	6278	15495	2902	10393	42741	9	5,6,8,9,11
7	1,2,3,6,11	1878	304	1910	74	2217	6383	3	4,5,8,9,10
7	1,3,4,8,9	30287	42693	6085	50075	3654	132794	1	2,5,6,10,11
7	1,3,6,8,11	614	550	194	152	297	1807	6	2,4,5,9,10
8	1,2,3,6,10	1410	2694	598	3109	4628	12439	9	4,5,7,9,11
8	1,2,5,6,7	44866	11396	12870	62375	5226	136733	1	3,4,9,10,11
8	1,2,5,6,9	9820	2570	1833	13526	2613	30362	2	3,4,7,10,11
8	1,2,5,10,11	2589	9511	2031	3027	22996	40154	1	3,4,6,7,9
8	1,2,9,10,11	723	6103	2031	3933	11728	24518	1	3,4,5,6,7
8	2,3,4,5,11	6507	1387	3142	3765	3887	18688	9	1,6,7,9,10
8	2,3,6,10,11	3114	328	109	3458	4230	11239	3	1,4,5,7,9
8	2,4,5,9,11	11222	3936	1011	12483	11060	39712	1	1,3,6,7,10
8	2,4,6,7,10	1990	104	1315	1442	3186	8037	18	1,3,5,9,11
8	2,7,9,10,11	1380	241	228	1214	2219	5282	3	1,3,4,5,6
9	1,3,5,6,11	19212	13094	38497	14620	12438	97861	1	2,4,7,8,10
9	1,5,6,7,10	1198	19329	4916	23140	2066	50649	1	2,3,4,8,11
9	2,3,4,5,7	1574	1628	1448	6753	3144	14547	12	1,6,8,10,11
9	2,3,4,5,10	14378	20588	3032	37755	37728	113481	1	1,6,7,8,11
9	2,3,5,7,11	290	104	4493	4616	2172	11675	8	1,4,6,8,10
9	3,5,7,8,10	1726	6667	3702	5104	2392	19591	1	1,2,4,6,11
10	1,4,5,6,9	14386	23140	33623	3752	9577	84478	1	2,3,7,8,11
10	1,4,5,7,8	14270	30691	47880	6133	10247	109221	1	2,3,6,9,11
10	1,4,5,9,11	4022	5844	9829	11	3752	23458	1	2,3,6,7,8
11	1,3,4,6,10	3149	4165	2999	16899	15406	42618	6	2,5,7,8,9
11	1,4,6,7,9	16	898	1756	853	2136	5659	13	2,3,5,8,10
11	3,4,6,7,10	608	1150	5848	3149	5748	16503	6	1,2,5,8,9
11	3,4,6,8,9	5074	16942	25719	14465	36606	98806	1	1,2,5,7,10
11	4,5,6,7,10	280	228	686	639	166	1999	16	1,2,3,8,9

APPENDIX B. THE CERTIFICATES

Table 6 lists the 28 relations of X_4, Tables 7 and 8 the 161 relations of X_3, Table 9 the 46 relations of Y_4, and Tables 10 to 12 the 282 relations of Y_3, each ordered by p and then by σ. The weights are the integers $m_j = M\lambda_j$ with $\gcd(m_0, \ldots, m_d) = 1$; the largest M is 136,733 for X_4, 261,355 for X_3, 185,165 for Y_4 and 189,499 for Y_3. To check a row, multiply each point x_{s_j} by m_j, add, and compare with Mx_p coordinate by coordinate; then compute one determinant of size $d+1$, whose absolute value must be gM.

TABLE 7. The 161 relations behind Theorem B. Each row gives $\sum_j m_j x_{s_j} = M x_p$ with $\sigma = \{s_1, \ldots, s_4\}$ in increasing order, positive integer weights m_j with sum M, and $|\det \sigma| = gM$ for the 4×4 determinant of the rows $(1, x_s)$.

p	σ	m_1	m_2	m_3	m_4	M	g	p	σ	m_1	m_2	m_3	m_4	M	g
6	1,2,3,12	1939	18838	7809	46242	74828	3	9	3,4,10,14	88	1	49	6	144	24
6	1,2,10,11	9628	26527	16482	12471	65108	16	9	3,4,14,15	383	192	25	98	698	12
6	2,3,4,8	34648	2701	12313	31010	80672	1	9	3,5,10,13	17959	6	10191	1152	29308	3
6	2,3,7,9	37155	8333	10899	22519	78906	1	9	3,7,10,12	25061	516	13953	306	39836	2
6	2,3,10,13	18392	4499	2471	15911	41273	2	9	3,8,10,13	3029	18	1725	168	4940	1
6	2,4,8,15	44768	15282	42383	2701	105134	1	10	2,4,5,9	843	67592	2067	6503	77005	8
6	2,4,9,14	22436	853	4834	22793	50916	1	10	2,4,5,14	688	155085	5760	13006	174539	4
6	2,4,11,13	23569	922	1769	29068	55328	6	10	3,4,6,11	102	2399	8	164	2673	78
6	2,5,7,13	11252	894	3044	10945	26135	4	10	4,5,6,7	66389	3431	678	5263	75761	4
6	2,7,8,12	6744	757	4793	13076	25370	2	10	4,5,6,15	58359	1525	1234	5263	66381	4
6	2,7,8,15	20536	7641	4885	13076	46138	2	10	4,5,8,12	15380	550	941	151	17022	16
6	2,8,9,13	9620	98	3784	8101	21603	1	10	4,5,11,13	15563	364	492	1144	17563	5
6	2,8,10,14	162	110	49	59	380	33	10	4,7,9,11	22967	156	1352	1668	26143	2
6	2,8,12,13	1388	590	1892	757	4627	2	10	4,8,11,15	17657	104	1076	1196	20033	2
6	2,8,12,14	1732	337	1617	1514	5200	1	10	4,11,14,15	3123	192	26	208	3549	8
6	2,9,13,15	2088	676	1843	98	4705	1	11	1,3,5,12	265	7257	36613	35711	79846	4
6	2,12,14,15	4536	1977	5036	337	11886	1	11	2,3,5,9	373	67	1019	829	2288	132
7	1,2,4,14	4432	26	8779	2337	15574	28	11	2,4,5,14	22503	6553	82862	62621	174539	4
7	1,3,4,10	678	10	151	1297	2136	8	11	2,5,6,9	541	13490	8844	6698	29573	1
7	1,4,6,9	75367	143525	6344	27906	253142	1	11	2,5,6,15	21	323	194	197	735	34
7	1,4,6,13	8006	15830	55	4651	28542	6	11	2,5,8,10	3511	13381	7879	3834	28605	28
7	1,4,10,13	30	29	33	8	100	10	11	2,5,12,13	3067	23571	10347	12688	49673	2
7	1,4,10,14	64	43	91	10	208	8	11	2,5,13,15	4723	14241	3298	10347	32609	2
7	1,4,10,15	152	44	278	5	479	16	11	3,5,6,7	1676	83567	49814	30595	165652	1
8	1,2,3,7	17633	28610	171431	22934	240608	2	11	3,5,8,12	826	4874	265	4683	10648	4
8	1,2,3,13	20655	26750	157849	45868	251122	1	11	4,5,6,15	238	39299	28782	20189	88508	3
8	1,2,3,15	11419	10214	51009	22934	95576	2	11	5,6,9,10	119571	88052	53191	541	261355	1
8	1,2,5,10	66745	3274	41539	49561	161119	12	11	5,6,10,15	103633	75898	714	53191	233436	1
8	1,3,6,9	2193	14585	5988	1030	23796	5	11	5,6,12,14	3785	1124	1749	1406	8064	5
8	1,3,6,12	2616	17890	5176	2575	28257	2	11	5,8,12,15	11307	2407	9968	826	24508	4
8	1,3,11,14	4523	2132	5932	9799	22386	4	12	1,2,4,5	3861	7214	8827	2875	22777	92
8	1,5,6,13	24028	16925	4483	61335	106771	4	12	1,2,9,10	467	5591	7309	2418	15785	28
8	1,5,7,11	3284	2091	4838	1964	12177	14	12	2,3,4,14	6064	3551	3779	4228	17622	4
8	1,5,12,14	4225	2788	1263	10679	18955	16	12	2,4,7,11	4201	1046	6290	2977	14514	26
8	1,7,11,15	646	205	748	861	2460	34	12	2,4,8,9	4886	2792	3343	3678	14699	4
8	1,11,13,15	9778	10804	6970	11603	39155	1	12	2,4,8,15	16672	12154	16385	7356	52567	2
9	1,3,4,5	1926	43091	24609	2082	71708	4	12	2,5,7,10	19372	6561	30758	4706	61397	10
9	1,3,10,15	69	5116	3003	474	8662	4	12	2,8,10,11	28610	35593	28094	1370	93667	4
9	3,4,5,14	14479	7377	294	1926	24076	4	12	2,9,10,13	17111	14023	6497	13076	50707	1
9	3,4,7,15	1844	906	75	639	3464	4	12	2,9,10,14	3527	2265	1514	3269	10575	4
9	3,4,8,11	1495	771	78	48	2392	32	12	2,10,13,15	26536	11836	29826	14023	82221	1

TABLE 8. The relations behind Theorem B, continued. Each row gives $\sum_j m_j x_{s_j} = M x_p$ with $\sigma = \{s_1, \ldots, s_4\}$ in increasing order, positive integer weights m_j with sum M, and $|\det \sigma| = gM$ for the 4×4 determinant of the rows $(1, x_s)$.

p	σ	m_1	m_2	m_3	m_4	M	g	p	σ	m_1	m_2	m_3	m_4	M	g
12	3,4,6,15	64	468	2782	399	3713	8	14	1,5,13,15	288	225	12462	1207	14182	4
12	4,5,6,7	8542	2774	55354	9091	75761	4	14	1,7,9,11	107	492	410	262	1271	78
12	4,5,6,13	9726	1279	54191	9091	74287	4	14	1,8,9,10	493	7390	1956	2944	12783	4
12	4,6,8,11	2174	9752	1251	416	13593	13	14	1,9,12,13	915	7412	1942	544	10813	1
12	4,6,9,11	1771	11336	1807	205	15119	9	14	1,10,12,15	861	714	654	4012	6241	28
12	4,6,13,15	1236	7742	128	1279	10385	4	14	2,3,7,15	1308	7224	8445	6181	23158	4
12	5,6,9,10	657	196364	29492	34842	261355	1	14	2,3,11,13	68	1839	446	25292	27645	4
12	6,7,10,11	96074	13359	17426	8176	135035	1	14	3,4,7,8	639	1351	584	4262	6836	4
12	6,7,10,15	3492	73	626	511	4702	16	14	3,5,7,12	9500	1213	12690	4103	27506	4
12	6,8,10,15	3370	146	688	365	4569	8	14	3,5,8,13	674	185	483	16388	17730	2
12	6,9,10,11	29872	4453	5313	219	39857	3	14	3,6,7,15	6366	2616	7263	5047	21292	2
13	1,2,4,14	5485	1118	12107	199326	218036	2	14	3,6,9,13	3	1	2	29	35	366
13	1,3,4,12	14660	62357	37121	29606	143744	2	14	3,8,10,13	143	526	185	1616	2470	2
13	1,4,8,9	2617	14795	35938	5246	58596	1	14	3,8,13,15	18	171	1686	185	2060	2
13	1,4,8,14	3	14	26	61	104	86	14	3,12,13,15	168	57	1638	61	1924	6
13	1,6,10,14	1111	433	2612	34331	38487	4	14	4,8,9,13	127	652	258	1580	2617	1
13	1,6,10,15	17930	9491	21574	68662	117657	2	14	5,7,8,15	153	1137	789	335	2414	8
13	1,7,9,11	7247	46781	25215	19895	99138	1	14	7,8,9,10	986	4766	1250	1217	8219	2
13	1,7,12,15	2236	7328	3305	17285	30154	4	14	7,8,9,11	4690	4173	2981	1217	13061	2
13	1,9,10,11	32261	37064	46781	28706	144812	1	14	7,8,12,15	1113	1281	153	1465	4012	8
13	2,3,7,11	5213	53574	110580	23621	192988	1	14	8,9,11,13	1133	1051	127	9380	11691	1
13	2,7,8,15	816	16327	12479	16516	46138	2	15	1,2,5,9	149	1025	2593	39087	42854	26
13	2,7,10,14	566	10230	707	102737	114240	1	15	1,5,9,12	102	1039	13976	1025	16142	26
13	3,5,7,12	8302	1144	14043	4017	27506	4	15	1,9,11,12	269	85895	13507	615	100286	2
13	3,7,8,10	28	95	291	100	514	52	15	2,3,4,11	949	59735	28841	14709	104234	4
13	3,7,12,15	86	111	49	44	290	104	15	2,5,9,14	927	2721	38973	1937	44558	2
13	4,7,12,14	58	1771	309	17496	19634	2	15	2,8,9,11	151	608	56698	8669	66126	2
13	5,6,7,14	71	149	1557	11252	13029	4	15	3,4,5,7	2303	1039	866	1894	6102	16
13	5,7,8,9	335	2307	1104	68	3814	8	15	3,4,5,13	9661	5505	2266	7576	25008	4
13	6,7,8,15	204	1783	1367	1780	5134	8	15	3,5,7,10	4024	1376	3031	2078	10509	8
13	6,7,14,15	45	333	2734	71	3183	4	15	3,5,8,10	6597	1129	3031	5225	15982	8
13	7,8,9,12	3839	4183	5296	1340	14658	2	15	3,5,9,12	34	83	1188	97	1402	78
13	7,8,11,14	1051	215	192	4504	5962	1	15	3,6,9,11	551	534	57097	8691	66873	1
13	7,11,12,14	1813	116	215	14582	16726	1	15	3,10,11,12	38783	21151	8684	1618	70236	2
14	1,2,7,15	7224	3180	17781	61453	89638	4	15	5,6,8,9	4157	3404	567	61068	69196	2
14	1,3,5,10	5297	144	3022	8242	16705	16	15	5,7,9,13	487	747	2788	394	4416	2
14	1,3,7,11	437	2665	7462	2802	13366	12	15	5,9,12,14	399	5040	309	442	6190	6
14	1,4,5,11	1731	2378	1066	32	5207	52	15	9,10,11,14	12492	3	2022	469	14986	2
14	1,4,6,15	18987	15234	11388	94354	139963	2	15	9,11,13,14	945	153	6	29	1133	1
14	1,5,7,9	5041	10218	39908	332	55499	2								

TABLE 9. The forty-six relations behind Theorem C in $\mathbb{R}^4$. In each row the point x_p satisfies $\sum_j m_j x_{s_j} = M x_p$ with $\sigma = \{s_1, \ldots, s_5\}$ listed in increasing order, the weights m_j are positive integers with sum M, and $|\det \sigma| = gM$ for the 5×5 determinant of the rows $(1, x_s)$, $s \in \sigma$. The last column lists the seven points not used by the relation; every four-element subset of $\{1, \ldots, 13\}$ lies in one of these sets.

p	σ	m_1	m_2	m_3	m_4	m_5	M	g	unused points
8	1,2,3,9,12	1065	23360	32113	19634	1985	78157	2	4,5,6,7,10,11,13
8	2,3,4,5,7	9823	12191	1729	1393	1807	26943	33	1,6,9,10,11,12,13
8	2,3,5,12,13	5224	6205	1085	546	1599	14659	6	1,4,6,7,9,10,11
8	2,3,9,10,11	1039	1367	565	218	108	3297	5	1,4,5,6,7,12,13
9	1,2,7,8,10	584	25560	35635	12157	50487	124423	1	3,4,5,6,11,12,13
9	2,3,5,6,10	30730	5384	38739	32805	9809	117467	3	1,4,7,8,11,12,13
9	2,3,7,10,11	673	113	785	1171	76	2818	21	1,4,5,6,8,12,13
9	2,4,5,7,12	1696	375	1009	891	2882	6853	38	1,3,6,8,10,11,13
9	2,5,6,8,13	8995	14013	10234	5124	3611	41977	3	1,3,4,7,10,11,12
9	2,7,8,11,12	9261	6002	499	15763	10039	41564	1	1,3,4,5,6,10,13
10	1,2,5,6,12	29352	6810	35371	39580	56485	167598	2	3,4,7,8,9,11,13
10	1,2,7,12,13	36023	4334	39646	73314	22703	176020	1	3,4,5,6,8,9,11
10	1,3,4,7,11	1114	4885	8801	973	31221	46994	2	2,5,6,8,9,12,13
10	1,3,5,6,9	18438	4996	22895	33055	19957	99341	5	2,4,7,8,11,12,13
10	1,3,5,8,13	2127	1416	5224	318	13085	22170	20	2,4,6,7,9,11,12
10	1,4,5,8,12	2158	5995	6094	2755	14915	31917	6	2,3,6,7,9,11,13
10	1,4,11,12,13	349	126	1592	1234	315	3616	3	2,3,5,6,7,8,9
10	1,6,7,9,12	15664	5931	14264	10135	25537	71531	1	2,3,4,5,8,11,13
10	1,6,8,11,12	2161	2079	315	8987	3721	17263	3	2,3,4,5,7,9,13
10	1,6,9,11,12	3832	3213	945	14264	7687	29941	1	2,3,4,5,7,8,13
10	3,4,5,9,13	13629	27615	24194	12498	8815	86751	1	1,2,6,7,8,11,12
11	1,2,3,6,7	4243	1170	1025	547	10331	17316	84	4,5,8,9,10,12,13
11	1,2,3,7,10	29589	8801	6829	78800	22974	146993	2	4,5,6,8,9,12,13
11	1,2,5,12,13	4791	4452	44731	21711	43349	119034	2	3,4,6,7,8,9,10
11	1,3,7,9,12	23024	3663	51512	27931	3659	109789	1	2,4,5,6,8,10,13
11	1,4,5,6,8	707	13603	36966	12587	15170	79033	3	2,3,7,9,10,12,13
11	1,7,8,9,13	3925	9542	2180	1306	639	17592	4	2,3,4,5,6,10,12
11	2,3,4,5,13	5589	8151	11380	58942	51975	136037	3	1,6,7,8,9,10,12
11	2,4,5,7,10	6465	3161	31118	19668	110187	170599	1	1,3,6,8,9,12,13
11	2,5,8,10,13	1041	17581	1773	14808	16490	51693	1	1,3,4,6,7,9,12
11	3,4,5,9,12	375	2092	4186	2835	2673	12161	8	1,2,6,7,8,10,13
11	4,5,6,9,10	1334	11555	2841	8025	19599	43354	1	1,2,3,7,8,12,13
11	5,7,9,10,13	580	7	296	805	419	2107	3	1,2,3,4,6,8,12
12	3,4,6,7,9	48374	40099	15048	60901	20743	185165	1	1,2,5,8,10,11,13
12	3,4,7,8,13	14921	8456	20570	1439	18181	63567	1	1,2,5,6,9,10,11
12	3,5,6,7,13	630	393	356	414	1460	3253	12	1,2,4,8,9,10,11
12	3,5,6,8,11	10914	6173	23822	1842	26376	69127	1	1,2,4,7,9,10,13
12	3,6,7,8,10	21456	30879	25431	5233	62308	145307	1	1,2,4,5,9,11,13
13	1,2,4,6,10	784	4731	10732	32501	34775	83523	2	3,5,7,8,9,11,12
13	1,4,6,7,9	13603	12020	61532	2530	38162	127847	1	2,3,5,8,10,11,12
13	1,4,6,9,12	1413	3510	8632	6192	2530	22277	1	2,3,5,7,8,10,11
13	2,4,6,8,10	6969	19428	50071	1568	56333	134369	1	1,3,5,7,9,11,12
13	2,4,6,11,12	462	2045	2119	893	2782	8301	9	1,3,5,7,8,9,10
13	4,6,7,8,11	109	123	24	66	67	389	101	1,2,3,5,9,10,12
13	4,6,7,8,12	21492	14921	8577	11979	6767	63736	1	1,2,3,5,9,10,11
13	4,6,8,9,11	5431	7857	2754	1212	5269	22523	2	1,2,3,5,7,10,12

TABLE 10. The 282 relations behind Theorem C in $\mathbb{R}^3$. Each row gives $\sum_j m_j x_{s_j} = M x_p$ with $\sigma = \{s_1, \ldots, s_4\}$ in increasing order, positive integer weights m_j with sum M, and $|\det \sigma| = gM$ for the 4×4 determinant of the rows $(1, x_s)$.

p	σ	m_1	m_2	m_3	m_4	M	g	p	σ	m_1	m_2	m_3	m_4	M	g
8	1,2,3,17	25144	37291	12680	104210	179325	2	10	3,11,14,15	7784	61588	13267	9956	92595	1
8	1,2,4,15	9971	5154	1105	41910	58140	2	11	1,2,4,18	1767	283	13125	9135	24310	6
8	1,2,6,15	8476	9019	2210	38520	58225	1	11	1,4,7,18	5077	34393	566	21386	61422	3
8	1,2,6,16	1132	1549	626	3852	7159	10	11	1,4,14,17	2936	15838	8430	1599	28803	6
8	1,2,7,14	8288	89082	7615	75685	180670	2	11	2,3,4,14	24805	3387	84531	66414	179137	3
8	1,2,10,15	11726	7779	2210	52150	73865	1	11	2,4,5,10	2278	16350	4353	41139	64120	3
8	1,2,11,17	3072	755	2536	4914	11277	10	11	2,4,6,9	3910	11913	5580	2493	23896	2
8	1,2,16,18	1832	3103	5170	3130	13235	2	11	2,4,6,14	10234	22488	10161	4986	47869	1
8	1,4,5,15	35851	12077	10308	66954	125190	1	11	2,4,10,13	1096	11241	24681	4353	41371	3
8	1,4,9,13	4432	3263	7059	3600	18354	4	11	3,4,5,12	19596	47796	17849	28179	113420	1
8	1,4,17,18	7807	3573	11836	65	23281	6	11	3,4,5,17	788	3406	78	3131	7403	9
8	1,5,11,16	15840	6208	12317	13533	47898	1	11	3,4,6,16	11026	83114	8610	82079	184829	2
8	1,6,11,12	23156	1723	19133	27798	71810	3	11	3,4,8,18	2573	61919	23714	35873	124079	2
8	1,10,11,13	14501	2717	10309	14371	41898	1	11	3,4,12,13	8002	22873	4517	17849	53241	1
8	2,3,5,7	84607	32954	47433	19099	184093	6	11	3,4,13,16	7018	21782	17220	4517	50537	1
8	2,3,7,12	14925	5278	3485	15811	39499	18	11	3,4,14,15	600	6173	1619	4961	13353	15
8	2,3,9,11	22493	930	77514	20884	121821	2	11	4,5,8,10	22926	4477	6834	45231	79468	1
8	2,3,9,14	55135	834	51458	41768	149195	1	11	4,6,8,10	16088	4477	10558	2802	33925	1
8	2,4,9,10	3752	883	12919	2480	20034	2	11	4,7,10,16	8292	13	12227	5808	26340	2
8	2,5,6,9	55531	2624	25458	48324	131937	1	11	4,8,10,18	9574	3537	5146	4477	22734	1
8	2,6,7,11	26368	11373	3186	899	41826	4	11	4,9,10,12	877	127	1565	315	2884	10
8	2,6,7,17	80065	35576	8752	1798	126191	2	11	4,10,12,16	926	1387	13	635	2961	2
8	2,9,10,18	1363	2018	434	883	4698	2	11	4,10,13,16	2357	3509	13	1644	7523	2
9	1,2,6,7	1109	11176	5310	3550	21145	25	11	4,10,15,17	242	354	116	47	759	16
9	1,2,7,10	5458	20912	14135	22125	62630	6	12	1,2,5,15	288	572	4165	4920	9945	11
9	1,3,7,8	2785	542	7518	54278	65123	11	12	1,2,13,17	23	592	4037	1259	5911	5
9	1,4,7,17	35203	17702	14370	38202	105477	2	12	1,4,5,9	2168	4114	17043	11706	35031	4
9	1,5,6,10	70613	47308	31350	6034	155305	2	12	2,3,9,13	16645	2161	9586	149167	177559	1
9	1,5,7,14	60466	35504	671	50289	146930	4	12	2,4,8,13	5597	138	12449	75282	93466	1
9	1,7,8,15	2243	1556	6040	5962	15801	1	12	2,5,6,17	2276	4129	660	5988	13053	2
9	1,7,10,16	3057	1000	194	7842	12093	16	12	2,5,7,10	4236	9441	1430	5218	20325	24
9	1,7,11,13	6287	2396	5536	4534	18753	18	12	2,5,8,16	46	571	338	444	1399	12
9	1,7,12,18	48154	5719	55761	44446	154080	2	12	2,5,9,11	3012	21107	14808	6952	45879	3
10	1,3,4,6	19703	62	45165	30268	95198	9	12	2,6,8,13	1985	138	3461	23316	28900	1
10	1,3,4,14	17758	9012	64166	68103	159039	4	12	2,7,13,16	11786	1031	88730	16082	117629	1
10	2,3,11,14	524	1020	8956	2847	13347	19	12	2,8,13,14	25	43	294	3	365	46
10	3,4,5,17	19018	16928	19319	11362	66627	1	12	2,9,11,13	6632	6789	2161	63321	78903	1
10	3,4,7,12	36276	84564	19801	48858	189499	2	12	3,4,5,8	1296	744	25771	22684	50495	5
10	3,4,9,14	13888	56920	35516	42301	148625	2	12	4,5,8,18	1848	32117	31964	3240	69169	2
10	3,4,12,18	23544	44810	38961	39602	146917	1	12	5,8,10,14	1909	1804	276	24	4013	16
10	3,4,13,18	22059	29278	38961	13993	104291	1	12	5,8,11,17	1607	1548	196	348	3699	6
10	3,4,14,17	29064	52956	19319	71032	172371	1	13	1,2,5,18	6	7669	34150	17615	59440	3
10	3,4,15,18	16065	46816	38961	44851	146693	1	13	1,5,7,11	8095	56259	4654	24497	93505	6
10	3,4,16,18	16011	41345	38961	36731	133048	1	13	1,5,7,16	1061	71992	1073	73491	147617	2
10	3,6,11,12	10658	6216	58754	15889	91517	2	13	1,5,14,15	518	18546	1274	14411	34749	2
10	3,6,11,16	7480	5743	54002	15889	83114	2	13	2,4,5,7	25146	40594	89135	27433	182308	3
10	3,6,11,17	9374	4355	57794	15889	87412	2	13	2,5,10,17	410	4807	483	6114	11814	5
10	3,8,11,18	9574	3121	85400	25743	123838	1	13	2,5,16,17	6	184	115	121	426	21
10	3,9,11,14	12368	9956	113840	24761	160925	1	13	3,5,6,8	5435	86505	4364	35653	131957	1
10	3,9,11,18	9657	3121	87029	24761	124568	1	13	3,5,8,18	687	10529	3840	1091	16147	4
10	3,11,12,14	2434	15178	2489	3108	23209	4	13	3,5,9,11	3156	62827	24015	9851	99849	1
10	3,11,13,18	4425	29278	3121	7171	43995	1	13	3,5,12,18	39	1193	1920	347	3499	8

TABLE 11. The relations behind Theorem C in $\mathbb{R}^3$, continued. Each row gives $\sum_j m_j x_{s_j} = M x_p$ with $\sigma = \{s_1, \ldots, s_4\}$ in increasing order, positive integer weights m_j with sum M, and $|\det \sigma| = gM$ for the 4×4 determinant of the rows $(1, x_s)$.

p	σ	m_1	m_2	m_3	m_4	M	g	p	σ	m_1	m_2	m_3	m_4	M	g
13	4,5,6,12	31	9674	1901	23797	35403	4	15	3,4,9,12	7914	3877	20021	50870	82682	1
13	4,5,9,14	6101	46849	24307	5211	82468	2	15	3,6,8,16	758	351	8898	47807	57814	1
13	5,6,7,15	80462	658	1036	73943	156099	2	15	3,8,10,17	495	16687	7253	23128	47563	1
13	5,6,8,16	10250	631	2952	5435	19268	1	15	3,8,11,13	7722	29074	974	40577	78347	1
13	5,8,9,10	16203	402	6194	4397	27196	1	15	3,8,12,14	2047	4819	13474	935	21275	2
13	5,8,11,18	13540	5895	916	3107	23458	3	15	4,5,6,9	27851	16859	7793	98822	151325	2
13	5,8,15,17	1322	28	913	546	2809	1	15	4,7,8,12	31226	15863	66430	13737	127256	1
13	5,9,14,15	7694	518	350	6101	14663	2	15	4,7,9,13	17886	3884	41348	9323	72441	2
13	5,10,12,17	249	26	205	268	748	10	15	4,8,13,18	1164	10013	4182	4553	19912	4
13	5,11,12,14	5777	76	11853	1842	19548	2	15	4,9,11,16	13	153	53	437	656	10
13	5,12,15,18	125	119	52	28	324	6	15	4,9,11,17	412	2832	1041	2185	6470	2
14	1,2,6,12	4982	13798	97900	38765	155445	1	15	5,6,8,10	18019	8894	67266	17663	111842	1
14	1,6,7,16	1624	54376	1165	48748	105913	4	15	5,7,9,11	11219	1164	52485	35047	99915	2
14	3,5,6,7	36092	67550	59528	10001	173171	4	15	5,8,10,14	4758	17908	4991	4447	32104	2
14	3,5,6,9	2	5	10	4	21	10001	15	6,8,10,16	499	5180	758	18019	24456	1
14	3,6,7,13	14448	51772	4419	54040	124679	5	15	6,9,11,12	1066	17021	10477	10327	38891	2
14	3,6,9,12	1254	16038	3272	10001	30565	5	15	7,8,11,13	7722	53992	36302	19385	117401	1
14	3,6,17,18	462	6609	5261	6060	18392	1	15	8,9,10,12	506	4174	3811	4465	12956	1
14	4,5,6,18	1758	4073	32032	94694	132557	1	15	8,11,17,18	2395	744	2208	607	5954	4
14	4,6,7,10	2184	4788	2915	6076	15963	8	16	1,3,4,5	1511	1794	493	5658	9456	69
14	4,6,8,13	135	19583	9402	5339	34459	5	16	1,3,5,11	14778	15966	53464	11339	95547	3
14	6,7,9,11	21476	6817	4312	22680	55285	2	16	1,3,12,13	833	4094	23113	4939	32979	1
14	6,7,15,17	19236	459	15960	2296	37951	2	16	1,4,7,12	8447	21365	9407	20447	59666	6
14	6,8,10,17	10878	3804	435	5210	20327	2	16	1,6,10,13	6929	2032	15785	27191	51937	4
14	6,8,12,16	2139	812	680	115	3746	2	16	1,6,11,17	3942	561	9643	30577	44723	4
14	6,9,10,13	3604	1480	805	1124	7013	12	16	1,9,10,13	27	127	184	250	588	64
14	6,10,13,18	7468	825	1948	17760	28001	1	16	1,9,11,17	213	561	1019	2747	4540	4
14	6,11,12,18	101	7	30	214	352	74	16	1,10,12,17	956	3319	2611	6369	13255	2
14	6,13,16,18	571	56	275	820	1722	3	16	1,10,13,18	11377	27433	46551	8128	93489	1
15	1,2,3,16	2639	1231	1785	94945	100600	3	16	2,3,4,17	6353	4459	2285	62977	76074	6
15	1,3,11,17	8372	1576	15497	49513	74958	3	16	2,4,5,14	10134	426	6037	17059	33656	9
15	1,3,12,18	7822	22501	140059	425	170807	1	16	2,5,9,10	610	5777	10218	9267	25872	5
15	1,4,6,13	17104	11454	10508	33827	72893	6	16	2,5,11,18	5394	4016	1423	11988	22821	6
15	1,4,14,17	1476	1265	1151	5709	9601	18	16	2,6,10,12	8564	8489	1174	16772	34999	2
15	1,5,6,11	46893	53513	23868	64129	188403	2	16	2,7,11,12	13754	12684	44365	23484	94287	2
15	1,7,10,12	19344	6017	69043	85248	179652	2	16	2,10,14,17	844	639	913	3258	5654	6
15	1,9,10,13	747	2011	3196	3454	9408	4	16	2,11,13,14	6862	915	8712	11392	27881	3
15	1,10,14,16	688	891	109	12804	14492	4	16	2,11,17,18	270	179	1004	456	1909	24
15	1,11,16,18	2720	2869	43717	1890	51196	1	16	3,4,9,13	9154	13563	50537	69700	142954	1
15	2,3,4,17	18121	12551	1177	120299	152148	3	16	3,5,8,18	9434	28290	26713	151	64588	1
15	2,3,5,18	15823	2745	11957	28926	59451	2	16	3,5,12,15	92	219	309	1604	2224	1
15	2,3,13,14	8543	241	6933	13061	28778	2	16	3,8,10,13	3973	12753	1813	26735	45274	1
15	2,4,12,14	37926	814	37647	64573	140960	1	16	3,8,12,13	11	17	37	35	100	49
15	2,4,17,18	20247	2816	32810	37653	93526	1	16	3,9,12,17	2166	856	15082	271	18375	1
15	2,6,9,16	3508	2597	1358	23495	30958	1	16	4,5,7,9	36541	15929	9539	70831	132840	2
15	2,6,13,18	2551	1578	2002	1273	7404	2	16	4,6,15,17	634	21	18226	6904	25785	2
15	2,7,10,13	29280	14267	64417	33528	141492	2	16	4,7,8,17	12286	4958	25090	28499	70833	1
15	2,7,11,12	39806	30471	99346	18951	188574	1	16	4,8,10,17	454	8530	4958	20099	34041	1
15	2,7,16,18	8621	56	51686	15827	76190	1	16	4,8,13,18	4948	32471	24346	17883	79648	1
15	2,8,14,16	437	441	728	3614	5220	1	16	4,9,13,18	10651	32471	20684	9154	72960	1
15	2,9,10,17	290	979	1067	1961	4297	11	16	4,9,14,17	3627	9381	778	13325	27111	2
15	2,11,14,17	4096	605	5676	6009	16386	3	16	4,12,15,18	515	5423	20025	2104	28067	1

TABLE 12. The relations behind Theorem C in $\mathbb{R}^3$, continued. Each row gives $\sum_j m_j x_{s_j} = M x_p$ with $\sigma = \{s_1, \ldots, s_4\}$ in increasing order, positive integer weights m_j with sum M, and $|\det \sigma| = gM$ for the 4×4 determinant of the rows $(1, x_s)$.

p	σ	m_1	m_2	m_3	m_4	M	g	p	σ	m_1	m_2	m_3	m_4	M	g
16	4,13,14,15	358	3555	784	21979	26676	2	17	7,8,10,13	7837	22272	25295	77868	133272	1
16	5,6,8,11	7408	3960	20862	4817	37047	4	17	7,8,12,14	207	840	8736	2453	12236	10
16	5,6,10,15	2590	347	1124	29572	33633	2	17	7,9,11,13	3376	10648	12345	30214	56583	2
16	6,8,10,12	2095	8564	3746	12228	26633	2	17	7,9,12,14	33	280	2884	807	4004	30
16	6,9,11,13	1031	19926	13366	15386	49709	2	17	7,10,12,18	427	1127	6606	1068	9228	2
16	7,9,10,12	581	12444	16003	24456	53484	2	17	7,12,16,18	5	193	23	64	285	98
16	7,11,12,15	35	163	275	1058	1531	26	17	7,13,14,16	1077	13272	1303	18060	33712	2
16	8,11,14,17	1374	657	224	2608	4863	6	17	8,9,13,18	1931	1132	8770	3175	15008	1
16	9,11,12,14	166	121	205	47	539	78	18	1,3,6,17	4598	3520	28469	40671	77258	4
16	10,12,15,17	95	143	956	269	1463	2	18	1,5,8,14	1238	1029	4234	31391	37892	2
16	10,13,15,18	590	2191	11377	276	14434	1	18	1,5,10,14	3224	2454	2117	35985	43780	4
16	11,13,15,17	7	6	108	31	152	20	18	1,6,8,13	2020	34736	25067	12602	74425	1
17	1,2,6,13	12834	14226	29555	130340	186955	1	18	1,6,10,12	20554	85404	15649	61191	182798	1
17	1,3,5,14	10532	9394	49523	10677	80126	3	18	1,9,14,16	37	71	1034	170	1312	13
17	1,5,6,15	4877	21741	6864	30647	64129	2	18	1,11,12,14	7474	2428	11257	103699	124858	1
17	1,5,10,18	18814	61869	21127	42130	143940	1	18	2,3,7,14	5578	68	1175	47897	54718	14
17	1,5,12,18	1456	3829	21127	11660	38072	1	18	2,6,7,11	12646	31077	11442	28487	83652	2
17	1,8,13,14	1725	5598	26529	8333	42185	1	18	2,6,7,12	38773	94240	5743	28487	167243	2
17	1,10,13,14	4836	2799	39072	10645	57352	2	18	2,6,7,16	11452	38095	2982	28487	81016	2
17	1,12,13,18	455	3833	3829	2960	11077	1	18	2,9,14,15	263	1234	8626	580	10703	1
17	2,5,7,14	17528	25312	2343	32797	77980	6	18	3,6,9,13	11321	58231	39607	51406	160565	1
17	2,7,12,18	1127	1477	66893	24743	94240	2	18	3,6,9,16	203	25226	12439	25703	63571	2
17	3,5,9,18	8755	39774	22793	1691	73013	1	18	3,7,11,14	7336	29803	78092	68785	184016	1
17	3,6,9,13	78	79	372	1360	1889	85	18	3,9,13,14	2053	11426	7095	58231	78805	1
17	3,7,11,12	1006	9113	21820	65663	97602	2	18	4,6,7,10	3140	4428	8937	26063	42568	3
17	3,7,13,15	5048	4594	78893	41335	129870	1	18	4,6,9,17	1507	10531	9053	4801	25892	4
17	3,7,13,16	2606	3518	48376	41335	95835	1	18	4,7,8,14	196	289	3066	13095	16646	6
17	4,5,7,15	5207	17736	5817	51917	80677	2	18	4,9,12,14	396	3518	173	14459	18546	4
17	4,7,13,15	631	693	4434	4579	10337	8	18	6,7,8,12	48878	947	38773	17687	106285	2
17	5,6,7,16	6643	579	1305	27016	35543	8	18	6,7,9,15	33023	1041	9091	41845	85000	1
17	5,7,10,16	785	188	193	2861	4027	24	18	6,7,10,11	1533	2467	6323	2355	12678	4
17	5,9,11,18	23522	24714	8755	13608	70599	1	18	6,8,9,12	12146	7769	1894	4566	26375	1
17	5,9,14,15	381	219	184	549	1333	22	18	6,9,14,15	263	271	1041	433	2008	1
17	6,7,8,13	5059	582	15090	32026	52757	5	18	7,8,14,16	277	4564	20807	588	26236	2
17	6,7,10,12	89	342	1015	3803	5249	24	18	7,10,13,14	1867	6923	252	5406	14448	4
17	6,7,12,15	4693	1349	37951	12180	56173	2	18	7,10,14,15	7	25	28	4	64	252
17	6,9,13,16	839	2214	6998	3315	13366	2	18	9,10,14,17	4260	1167	19517	928	25872	1

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